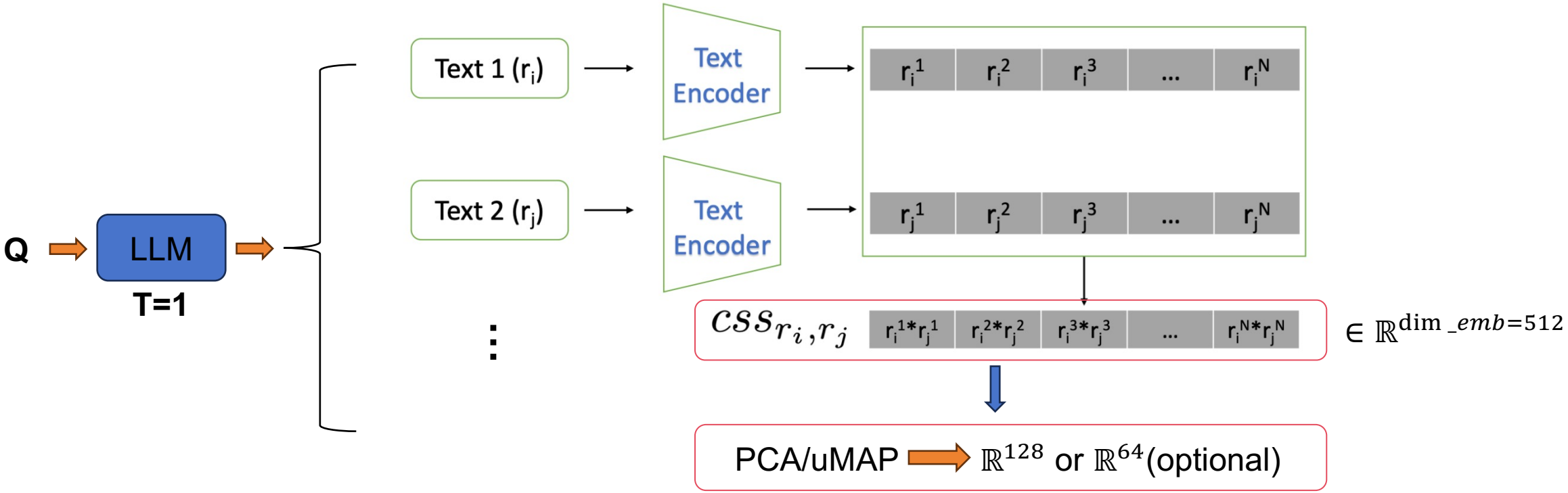
CSS

CSS[3]



Core Idea:

- to estimate uncertainty for black-box LLMs by leveraging CLIP-based contrastive similarity features.

$$U_{\text{Ecc}}^{css} = \left\| \left[\mathbf{e}^{\mathbf{css}'}{}_1^{\top}, \ldots, \mathbf{e}^{\mathbf{css}'}{}_m^{\top} \right] \right\|_2$$

$$U_{\text{Eig}}^{css} = \sum_{k=1}^{m} \max\left(0, 1 - \lambda_k^{css}\right)$$

$$U_{\text{Deg}}^{css} = \text{trace}(m - D^{css}) / m^2$$

[3] CSS: Contrastive Semantic Similarity for Uncertainty Quantification of LLMs, arxiv 2024

CSS[3]

- **Task Type:** Open-ended generation

- **Introduction:** Contrastive Semantic Similarity (CSS) extends semantic clustering by extracting CLIP-based text-pair features and applying spectral graph Laplacian clustering to estimate uncertainty. It quantifies decoding uncertainty in black-box LLMs, providing a semantic-consistency signal for selective generation and evaluation.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	Black-box	Post-hoc	Medium	Ranking

[3] CSS: Contrastive Semantic Similarity for Uncertainty Quantification of LLMs, arxiv 2024